

Functions: Composite & Inverse



Domain and range

- 1 Find the domain of $f(x) = \frac{1}{x-5}$.
 - 2 Find the domain of $g(x) = \sqrt{x-3}$.
 - 3 Find the domain of $h(x) = \frac{1}{\sqrt{x-1}}$.
 - 4 Find the range of $f(x) = x^2 + 3$, for all real x .
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Evaluating functions

- 5 If $f(x) = 2x^2 - 3x + 1$, find $f(-2)$.
 - 6 If $f(x) = x^3 - 4x + 2$, find $f(3)$.
 - 7 If $g(x) = \frac{x+3}{x-2}$, find $g(4)$.
 - 8 If $h(x) = \sqrt{2x+7}$, find $h(9)$.
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Composite functions

- 9 Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $fg(x)$ (meaning $f(g(x))$).
 - 10 Using the same f and g , find $gf(x)$, and explain why $fg(x) \neq gf(x)$ in general.
 - 11 Using the same $f(x) = 2x + 1$, find $ff(x)$ (meaning $f(f(x))$).
 - 12 Given $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, find $fg(x)$ (meaning $f(g(x))$).
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Finding an unknown in a composite function

- 13 Given $f(x) = 3x - 2$ and $g(x) = x + 4$, find the value of x such that $fg(x) = 13$.
- 14 Using the same $f(x) = 3x - 2$ and $g(x) = x + 4$, find the value of x such that $gf(x) = 15$.
- 15 Given $f(x) = 2x + 5$ and $g(x) = 3x - 1$, find the value of x such that $fg(x) = 21$.

16 Given $f(x) = x + 6$ and $g(x) = 2x$:

a Find $fg(x)$.

b Solve $fg(x) = 20$ for x .

Inverse functions

17 Find $f^{-1}(x)$ for $f(x) = \frac{2x+5}{3}$.

18 Find $f^{-1}(x)$ for $f(x) = x^2 + 4$, for $x \geq 0$, and explain why the restriction $x \geq 0$ is needed.

19 Find $f^{-1}(x)$ for $f(x) = \frac{4x-1}{5}$.

20 Find $f^{-1}(x)$ for $f(x) = \frac{x-7}{2}$.

The relationship between a function and its inverse graphically

21 If the point $(3, 11)$ lies on the graph of $y = f(x)$, state a point that must lie on the graph of $y = f^{-1}(x)$, and explain the general geometric relationship between the two graphs.

22 The graph of $y = f(x)$ passes through $(-2, 7)$. State the point that must lie on the graph of $y = f^{-1}(x)$.

23 The graph of $y = f(x)$ has a y -intercept at $(0, 9)$. State the corresponding point that must lie on the graph of $y = f^{-1}(x)$.

Range of a composite function

24 Given $f(x) = x^2$ with domain all reals, and $g(x) = x - 1$, find the range of $fg(x)$.

25 Given $f(x) = \sqrt{x}$ with domain $x \geq 0$, and $g(x) = x + 5$, find the range of $fg(x)$.

26 Given $f(x) = \frac{1}{x}$ with domain $x \neq 0$, and $g(x) = x - 2$, find the range of $fg(x)$.

27 Given $f(x) = x^2$ with domain all reals, and $g(x) = 3 - x$:

a Find $fg(x)$.

b State the range of $fg(x)$.

A well-designed word problem: currency conversion functions

- 28** This question has two parts. A currency exchange converts US dollars to euros using $f(x) = 0.92x$, then charges a flat 3-euro service fee, giving the final amount $g(f(x))$ where $g(y) = y - 3$. (a) Find the amount of euros received for 200 dollars. (b) Find the inverse function of $h(x) = g(f(x))$, and use it to determine how many dollars must be exchanged to receive exactly 100 euros.